

Aidan Manternach

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EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO
M.S. in Data Science and Machine Learning | GPA: 4.0
B.S. in Computer Engineering | GPA: 4.0

San Diego, CA
Expected June 2027
June 2025

EXPERIENCE

American Express | *Software Engineer Intern* June 2025 – August 2025

- Built a real-time anomaly detection system in **Python** leveraging **Kafka**, **PySpark**, and **SARIMAX** models to identify irregular patterns within a **continuous stream of data, ~7M events per hour**
- Automated the data processing and monitoring workflow with **Apache Airflow**, enabling seamless orchestration, proactive anomaly alerts, and improved system resilience across streaming pipelines

American Express | *Software Engineer Intern* June 2024 – August 2024

- Developed an automated **feature drift detection data batch** job pipeline using **SQL**, **PySpark**, and **Python** to track data distribution shifts across multiple time horizons and model inputs
- Implemented Airflow-based scheduling and visualization dashboards with integrated email alerts, providing continuous data quality insights and early detection of model performance degradation

Extreme Waves Inc. | *Software Engineer Intern* January 2024 – April 2024

- Developed a code library to automate testing workflows for phased array devices, reducing manual test time by 50% through direct firmware control and robust debugging features
- Developed a customer-facing **C#** GUI for device control with a senior engineer, enabling intuitive device configuration, real-time monitoring, and operational control

UCSD CS Department | *Instructional Assistant* September 2023 – December 2024

- Conducted 150+ one-on-one tutoring sessions, teaching core topics in **object-oriented design**, control structures, and functions, while administering and grading assignments, projects, and exams

HACKATHONS

UCSD SMASH - NSF HDR | *1st Place* January 2026

- Out-of-distribution neural forecasting model with **adaptive graph learning and temporal transformers** for multivariate time series; **7th globally** on the leaderboard

AI + Blockchain Hosted By AO Computing | *1st Place*

PROJECTS

VidShare | *Go, gRPC, HTTP, SQLite, FFmpeg, REST API*

- Built a scalable video storage and streaming service in **Go** using **gRPC microservices** and **consistent hashing**, achieving 100% availability and sub-10ms metadata lookups with **SQLite**
- Integrated **FFmpeg** adaptive bitrate encoding with **HTTP streaming**, cutting playback buffering by 35% and enabling efficient, scalable video delivery with optimized bandwidth utilization

TECHNICAL SKILLS

Programming Languages: Java, Python, Go, C/C++, SQL, JavaScript, Bash

Frameworks & Libraries: React, FastAPI, Node.js, Mockito, PyTorch, TensorFlow, PySpark

Tools: Docker, Kubernetes, AWS, Git, GitHub, Linux, MySQL, MongoDB

Coursework & Concepts: Computer Science, Data Structures & Algorithms, Database Systems, Cloud Infrastructure, Machine Learning, REST API, Network Systems, Computer Security